

Ass. 3

1) If $\{x_n\}$ is a seq. in (X, d) s.t. $x_n \rightarrow x_0$, then $\{x_0, x_1, x_2, \dots\}$ is compact.

2) prove of dis prove

a) $]0, 2]$ is not compact.

b) A seq. $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\} \subset \mathbb{R}$ is not compact.

3) prove that finite union of compact sets is also compact, but infinite union of compact sets need not be compact.

4) Show that in each compact metric space X $\exists$ a countable set E such that $\overline{E} = X$. $\overline{E} := E \cup E'$

5) All missing proofs in lecture 3-2. limit points of E